

Maths - Question 1

Original paper page 2; original question and option regions

1. The domain of the function $f(x) = \sqrt{\log_{0.5} x!}$

1) $\{0, 1, 2, 3, \dots\}$

2) $\{1, 2, 3, \dots\}$

3) $(0, \infty)$

4) $\{0, 1\}$

Key: 4

Sol: $0 < x! \leq 1$

$$x = 0, 1$$

Maths - Question 2

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2. If $f(x) = |x-1| + |x-2| + |x-3|$, $2 < x < 3$, then f is

1) an onto function but no one-one

2) one-one function but not onto

3) a bijection

4) neither one-one nor onto

Key: 3

Sol: $f(x) = x-1 + x-2 + 3-x$

$$f(x) = x$$

Maths - Question 3

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3. The greatest positive integer which divides $(n+16)(n+17)(n+18)(n+19)$, for all positive integers n , is

1) 6

2) 24

3) 28

4) 20

Key: 2

Sol: ${}^n P_r$ is divisible by $r!$.

$$\begin{vmatrix} a & b & c \\ , & & \end{vmatrix}$$

Maths - Question 4

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4. If a, b, c are distinct positive real numbers, then the value of the determinant $\begin{vmatrix} b & c & a \\ c & a & b \end{vmatrix}$ is

1) < 0

2) > 0

3) 0

4) ≥ 0

Key: 1

Sol: $\Delta = 3abc - a^3 - b^3 - c^3$

$$= -(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

Maths - Question 5

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5. If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in geometric progression with the same common ratio, then the points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are

1) vertices of an equilateral triangle

2) vertices of a right angled triangle

3) vertices of a right angled isosceles triangle

4) collinear

Key: 4

Sol: $(x_1, y_1)(x_1r, y_1r)(x_1r^2, y_1r^2)$

$$\frac{y_1(r-1)}{x(r-1)} = \frac{y_1r(r-1)}{x_1r(r-1)}$$

collinear

Maths - Question 6

Original paper page 3; original question and option regions

6. The equations $x - y + 2z = 4$

$$3x + y + 4z = 6$$

$$x + y + z = 1 \text{ have}$$

1) unique solution

3) no solution

Key: 2

Sol: $4x + 6z = 10$

$$2x + 3z = 5$$

$$\frac{4}{2} = \frac{6}{3} = \frac{10}{5}$$

2) infinitely many solutions

4) two solutions

Maths - Question 7

Original paper page 3; original question and option regions

7. The locus of the point representing the complex number z for which $|z + 3|^2 - |z - 3|^2 = 15$ is

1) a circle

2) a parabola

3) a straight line

4) an ellipse

Key: 3

$$\text{Sol: } (x+3)^2 + y^2 - [(x-3)^2 + y^2] = 15$$

$$12x = 15$$

$$x = \frac{5}{4}$$

$$(1+i)^{2016}$$

Maths - Question 8

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8. $\frac{1}{(1-i)^{2014}} =$

1) $-2i$

2) $2i$

3) 2

4) -2

Key: 1

Sol: $\left(\frac{1+i}{1-i}\right)^{2014} (1+i)^2$

$$= i^{2014} \cdot (2i)$$

$$= -2i$$

Maths - Question 9

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9. If $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is

1) 3

2) 4

3) 8

4) 2

Key: 4

$$\text{Sol: } |z_1| = 1 \Rightarrow \bar{z}_1 = \frac{1}{z_1}$$

$$|z_2| = 2 \Rightarrow \bar{z}_2 = \frac{4}{z_2}$$

$$|z_3| = 3 \Rightarrow \bar{z}_3 = \frac{9}{z_3}$$

$$|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$$

$$|z_1z_2z_3| \left| \frac{9}{z_3} + \frac{4}{z_2} + \frac{1}{z_1} \right| = 12$$

$$|\bar{z}_3 + \bar{z}_2 + \bar{z}_1| = 2$$

$$|z_1 + z_2 + z_3| = 2$$

Maths - Question 10

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10. If $1, z_1, z_2, \dots, z_{n-1}$ are the n^{th} roots of unity, then $(1 - z_1)(1 - z_2) \dots (1 - z_{n-1}) =$

1) 0

2) $n-1$

3) n

4) 1

Key: 3

$$\text{Sol: } \frac{x^n - 1}{x - 1} = (x - z_1)(x - z_2) \dots (x - z_{n-1})$$

$$\lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1} = (1 - z_1)(1 - z_2) \dots (1 - z_{n-1})$$

$$n = (1 - z_1)(1 - z_2) \dots (1 - z_{n-1})$$

Maths - Question 11

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11. If $12^{4+2x^2} = (24\sqrt{3})^{5x^2-2}$, then $x =$

1) $\pm\sqrt{\frac{13}{12}}$

2) $\pm\sqrt{\frac{14}{5}}$

3) $\pm\sqrt{\frac{12}{13}}$

4) $\pm\sqrt{\frac{5}{14}}$

Key: 2

Sol: $12^{4+2x^2} = (24\sqrt{3})^{3x^2-2}$

$$(2\sqrt{3})^{8+4x^2} = (2\sqrt{3})^{9x^2-6}$$

$$8+4x^2 = 9x^2-6$$

$$5x^2 = 14$$

$$x = \pm\sqrt{\frac{14}{5}}$$

Maths - Question 12

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12. The product and sum of the roots of the equation $|x^2| - 5|x| - 24 = 0$ are respectively

1) $-64, 0$

2) $-24, 5$

3) $5, -24$

4) $0, 72$

Key: 1

Sol: $|x|^2 - 5|x| - 24 = 0$

$$(|x| - 8)(|x| + 3) = 0$$

$$|x| = 8 \Rightarrow x = \pm 8$$

$$\alpha\beta = -64$$

$$\lambda + \beta = 0$$

Maths - Question 13

Original paper page 4; original question and option regions

13. The number of real roots of the equation $x^5 + 3x^3 + 4x + 30 = 0$ is

1) 1

2) 2

3) 3

4) 5

Key: 1

Sol: $f(x) = x^5 + 3x^3 + 4x + 30 = 0$

At most zero positive roots

$$f(-x) = -x^5 + 3x^3 - 4x + 36$$

Maths - Question 14

Original paper page 5; original question and option regions

14. If the coefficients of the equation whose roots are k times the roots of the equation

$x^3 + \frac{1}{4}x^2 - \frac{1}{16}x + \frac{1}{144} = 0$, are integers then a possible value of k is

1) 3

2) 12

3) 9

4) 4

Key: 2

Sol: $f(x) = x^3 + \frac{1}{4}x^2 - \frac{1}{16}x + \frac{1}{144} = 0$

$$f\left(\frac{x}{m}\right) = x^3 + \frac{1}{4}mx^2 - \frac{1}{16}m^2x + \frac{m^3}{144} = 0$$

$$4 = 2^2$$

$$16 = 2^4$$

$$144 = 2^4 > 3^2$$

$$m = 12$$

Maths - Question 15

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15. The sum of all 4-digit numbers that can be formed using the digits 2, 3, 4, 5, 6 without repetition, is

1) 533820

2) 532280

3) 533280

4) 532380

Key: 3

Sol: ${}^4P_3(1111)(2+3+4+5+6)$

$= 5,33,280$

Maths - Question 16

Original paper page 5; original question and option regions

16. If a set A has 5 elements, then the number of ways of selecting two subsets P and Q from A such that P and Q are mutually disjoint, is

1) 64

2) 128

3) 243

4) 729

Key: 3

Sol:

$$x \in P, 2 \in Q$$

$$x \in P, 4 \notin Q$$

$$x \notin P, 4 \in Q$$

$$x \notin P, 2 \in Q$$

$$A \cap B = \phi$$

$$3^5 = 243$$

Maths - Question 17

Original paper page 5; original question and option regions

17. The coefficient of x^4 in the expansion of $(1 + x + x^2 - x^3)^4$ is

1) 31

2) 30

3) 25

4) -14

Key: 1

Maths - Question 18

Original paper page 6; original question and option regions

18. If the middle term in the expansion of $(1+x)^{2n}$ is the greatest term, then x lies in the interval

1) $\left(\frac{n}{n+1}, \frac{n+1}{n}\right)$

2) $\left(\frac{n+1}{n}, \frac{n}{n+1}\right)$

3) $(n-2, n)$

4) $(n-1, n)$

Key: 1

Sol: $T_n < T_{n+1} > T_{n+2}$

$${}^{2n}C_{n-1}x^{n-1} < {}^{2n}C_n x^n > {}^{2n}C_{n+1}x^{n+1}$$

$$\frac{{}^{2n}C_{n-1}}{{}^{2n}C_n} < x \quad \left| \quad \frac{{}^{2n}C_n}{{}^{2n}C_{n+1}} > x \right.$$

$$\frac{n}{n+1} < n \quad \left| \quad \frac{n+1}{n} > n \right.$$

$$\boxed{\frac{n}{n+1} < n < \frac{n+1}{n}}$$

Maths - Question 19

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19. To find the coefficient of x^4 in the expansion of $\frac{3x}{(x-2)(x-1)}$, then interval in which the expansion is valid, is

1) $-2 < x < \infty$

2) $-\frac{1}{2} < x < \frac{1}{2}$

3) $-1 < x < 1$

4) $-\infty < x < \infty$

Key: 3

$$\text{Sol: } \frac{3x}{(x-2)(x-1)} = \frac{6}{x-2} + \frac{-3}{x-1}$$

$$= 3(1-x)^{-1} - 3\left(1-\frac{x}{2}\right)^{-1}$$

$$|x| < 1 \& \left|\frac{x}{2}\right| < 1$$

$$\& |x| < 2$$

$$\boxed{-1 < x < 1}$$

Maths - Question 20

Original paper page 6; original question and option regions

20. If $(1 + \tan \alpha)(1 + \tan 4\alpha) = 2$, $\alpha \in \left(0, \frac{\pi}{16}\right)$, then $\alpha =$

1) $\frac{\pi}{20}$

2) $\frac{\pi}{30}$

3) $\frac{\pi}{40}$

4) $\frac{\pi}{60}$



Maths - Question 21

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21. If $\cos \theta = \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta}$, then one of the value of $\tan\left(\frac{\theta}{2}\right)$ is

1) $\cot \frac{\beta}{2} \tan \frac{\alpha}{2}$

2) $\tan \alpha \tan \frac{\beta}{2}$

3) $\tan \frac{\beta}{2} \cot \frac{\alpha}{2}$

4) $\tan n^2 \frac{\alpha}{2} \tan^2 \frac{\beta}{2}$

Key: 1

Sol: $\cos \theta = \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta}$

$$\tan \frac{\theta}{2} = \sqrt{\frac{1 - \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta}}{1 + \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta}}}$$

$$= \sqrt{\frac{1 - \cos \alpha \cos \beta - \cos \alpha + \cos \beta}{1 + \cos \alpha \cos \beta + \cos \alpha - \cos \beta}}$$

$$= \sqrt{\frac{(1 + \cos \beta)(1 - \cos \alpha)}{(1 - \cos \beta)(1 + \cos \alpha)}}$$

$$= \sqrt{\cot^2 \frac{\beta}{2} \tan^2 \frac{\alpha}{2}}$$

$$= \cot \frac{\beta}{2} \tan \frac{\alpha}{2}$$

Maths - Question 22

Original paper page 7; original question and option regions

22. The value of the expression

$$\frac{1 + \sin 2\alpha}{\cos(2\alpha - 2\pi) \tan\left(\alpha - \frac{3\pi}{4}\right)} - \frac{1}{4} \sin 2\alpha \left(\cot \frac{\alpha}{2} + \cot \left(\frac{3\pi}{2} + \frac{\alpha}{2} \right) \right) \text{ is}$$

1) 0

2) 2

3) $\sin^2 \frac{\alpha}{2}$

4) $\sin^2 \alpha$

Maths - Question 23

Original paper page 8; original question and option regions

23. If $\frac{1}{6}\sin\theta, \cos\theta$ and $\tan\theta$ are in geometric progression, then the solution set of θ is

- 1) $2n\pi \pm \left(\frac{\pi}{6}\right)$ 2) $2n\pi \pm \left(\frac{\pi}{3}\right)$ 3) $n\pi + (-1)^n \left(\frac{\pi}{3}\right)$ 4) $n\pi + \left(\frac{\pi}{3}\right)$

Key: 2

$$\text{Sol: } \cos^2 \theta = \frac{1}{6} \sin \theta \frac{\sin \theta}{\cos \theta}$$

$$6\cos^3 \theta = \sin^2 \theta$$

$$6\cos^3 \theta + \cos^2 \theta - 1 = 0$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 2n\pi \pm \frac{\pi}{3}$$

$$\left(\cos \theta - \frac{1}{2}\right)(6\cos^2 \theta + 4\cos \theta + 2)$$

$$\Delta < 0$$

Maths - Question 24

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24. If $x = \sin(2 \tan^{-1} 2)$ and $y = \sin\left(\frac{1}{2} \tan^{-1} \frac{4}{3}\right)$, then

1) $x > y$

2) $x = y$

3) $x = 0 = y$

4) $x < y$

Key: 1

Sol:

$$x = \sin(2 \tan^{-1}(2)) \quad y = \sin\left(\frac{1}{2} \tan^{-1} \frac{4}{3}\right)$$

Maths - Question 25

Original paper page 9; original question and option regions

25. If $\cosh(x) = \frac{5}{4}$, then $\cosh(3x) =$

1) $\frac{61}{16}$

2) $\frac{63}{16}$

3) $\frac{65}{16}$

4) $\frac{61}{63}$

Key: 3

Sol: $\cosh(x) = \frac{5}{4}$

$$\cosh(3x) = 4\cosh^3(x) - 3\cosh(x)$$

$$= 4\left(\frac{125}{64}\right) - 3\left(\frac{5}{4}\right)$$

$$= \frac{125 - 60}{16} = \frac{65}{16}$$

$$(B - C) \cap A$$

$$(C - A) \cap B$$

$$(A - B) \cap C$$

Maths - Question 26

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26. In $\triangle ABC$, if $x = \tan\left(\frac{B}{2}\right)\tan\frac{C}{2}$, $y = \tan\left(\frac{C}{2}\right)\tan\frac{A}{2}$ and $z = \tan\left(\frac{A}{2}\right)\tan\frac{B}{2}$, then

$$(x + y + z) =$$

1) xyz

2) $-xyz$

3) $2xyz$

4) $\frac{1}{2}xyz$

Key: 2

Maths - Question 27

Original paper page 10; original question and option regions

27. In $\triangle ABC$, if the sides a, b, c are in geometric progression and the largest angle exceeds the smallest angle by 60° , then $\cos B =$

1) $\frac{\sqrt{13}+1}{4}$

2) $\frac{1-\sqrt{13}}{4}$

3) 1

4) $\frac{\sqrt{13}-1}{4}$

Key: 4

Sol: $b^2 = ac$

$C - A = 60^\circ$

$\sin^2 B = \sin A \sin C$

$$\sin^2 B = \frac{1}{2} [\cos(C - A) - \cos(C + A)]$$

$$2 \sin^2 B = \frac{1}{2} - \sin B$$

Maths - Question 28

Original paper page 11; original question and option regions

28. In a $\triangle ABC$ if $\angle A = 90^\circ$, then $\cos^{-1}\left(\frac{r_2}{r_2 + r_3}\right)$ is equal to

1) 90°

2) 30°

3) 60°

4) 45°

Key: 3

Sol: $\angle A = 90^\circ$

$$r_2 + r_3 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} + 4R \sin \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}$$

$$= 4R \sin \frac{A}{2} \left[\sin \frac{B+C}{2} \right]$$

$$= 4R \sin \frac{A}{2} \cos \frac{A}{2}$$

$$= 2R \sin A$$

$$= 2R$$

$$\frac{R}{r_2 + r_3} = \frac{1}{2}$$

$$\cos^{-1}\left(\frac{R}{r_2 + r_3}\right) = \frac{\pi}{3}$$

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Maths - Question 29

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Maths - Question 29 (continued)

Original paper page 2; original question and option regions

ANDHRA PRADESH

MATHS

Maths - Question 30

Original paper page 12; original question and option regions

30. For three vectors $\vec{p}, \vec{q}$ and $\vec{r}$, if $\vec{r} = 3\vec{p} + 4\vec{q}$ and $2\vec{r} = \vec{p} - 3\vec{q}$, then

- 1) $|\vec{r}| < 2|\vec{q}|$ and $\vec{r}, \vec{q}$ have the same direction 2) $|\vec{r}| > 2|\vec{q}|$ and $\vec{r}, \vec{q}$ have opposite directions
 3) $|\vec{r}| < 2|\vec{q}|$ and $\vec{r}, \vec{q}$ have opposite directions 4) $|\vec{r}| > 2|\vec{q}|$ and $\vec{r}, \vec{q}$ have the same direction

Key: 2

$$\text{Sol: } \hat{r} = 3\hat{p} + 4\hat{q}$$

$$2\hat{r} = \hat{p} - 3\hat{q}$$

$$\hat{r} = 3\hat{p} + 4\hat{q}$$

$$6\hat{r} = 3\hat{p} - 9\hat{q}$$

$$5\hat{r} = -13\hat{q}$$

$$|\hat{r}| = \frac{13}{5}|\hat{q}|$$

$$|\hat{r}| > 2|\hat{q}|$$

Maths - Question 31

Original paper page 12; original question and option regions

31. If $\vec{a} = 2\vec{i} + 3\vec{j} - 5\vec{k}$, $\vec{b} = m\vec{i} + n\vec{j} + 12\vec{k}$ and $\vec{a} \times \vec{b} = \vec{0}$, then $(m, n) =$

1) $\left(\frac{-24}{5}, \frac{-36}{5}\right)$

2) $\left(\frac{-24}{5}, \frac{36}{5}\right)$

3) $\left(\frac{24}{5}, \frac{-36}{5}\right)$

4) $\left(\frac{24}{5}, \frac{36}{5}\right)$

Key: 1

Sol: $\frac{m}{2} = \frac{n}{3} = \frac{12}{-5}$

$$m = -\frac{24}{5}, n = -\frac{36}{5}$$

Maths - Question 32

Original paper page 12; original question and option regions

32. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and the angle between $\vec{a}$ and $\vec{b}$ is 120° , then $|4\vec{a} + 3\vec{b}|$ is equal to

1) 25

2) 7

3) 13

4) 12

Key: 4

$$\text{Sol: } |\vec{a}| = 3, |\vec{b}| = 4, (\vec{a} \cdot \vec{b}) = 12 \cos 120^\circ = -6$$

$$|4\vec{a} + 3\vec{b}| = \sqrt{16a^2 + 9b^2 + 24\vec{a} \cdot \vec{b}}$$

$$= \sqrt{16(9) + 9(16) + 24(3)(4)(-1/2)}$$

$$= \sqrt{288 - 144}$$

$$= \sqrt{144}$$

$$= 12$$

Maths - Question 33

Original paper page 13; original question and option regions

33. If $\vec{a}, \vec{b}, \vec{c}$ are non-zero vectors such that $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3} |\vec{b}| |\vec{c}| \vec{a}$ and θ is the angle between the vectors $\vec{b}, \vec{c}$ then $\sin \theta =$

1) $\frac{2\sqrt{2}}{3}$

2) $\frac{1}{3}$

3) $\frac{\sqrt{2}}{3}$

4) $\frac{2}{3}$

Key: 1

$$\text{Sol: } (\hat{a} \times \hat{b}) \times \hat{c} = \frac{1}{3} |\hat{b}| |\hat{c}| \hat{a}$$

$$-(\hat{b} \cdot \hat{c}) \hat{a} = \frac{1}{3} |\hat{b}| |\hat{c}| \hat{a} \quad \hat{c} \perp \hat{a}$$

$$(\hat{b}, \hat{c}) = \theta$$

$$-|\hat{b}| |\hat{c}| \cos \theta \hat{a} = \frac{1}{3} |\hat{b}| |\hat{c}| \hat{a}$$

$$\cos \theta = -\frac{1}{3}$$

$$\sin \theta = \sqrt{1 - \frac{1}{9}}$$

$$= \frac{2\sqrt{2}}{3}$$

Maths - Question 34

Original paper page 13; original question and option regions

34. If $a(\bar{\alpha} \times \bar{\beta}) + b(\bar{\beta} \times \bar{\gamma}) + c(\bar{\gamma} \times \bar{\alpha}) = \bar{0}$ and atleast one of the scalars a, b, c is non zero then the vectors $\bar{\alpha}, \bar{\beta}, \bar{\gamma}$ are

1) parallel

2) non coplanar

3) coplanar

4) mutually perpendicular

Key: 3

Sol: $a \neq 0$

$$a(\hat{\alpha} \times \hat{\beta}) \times \hat{\gamma}$$

$$[\hat{\alpha} \hat{\beta} \hat{\gamma}] = 0$$

$\hat{\alpha}, \hat{\beta}, \hat{\gamma}$ coplanar

Maths - Question 35

Original paper page 13; original question and option regions

35. If the mean of 10 observations is 50 and the sum of the squares of the deviations of the observations from the mean is 250, then the coefficient of variation of those observations is

1) 25

2) 50

3) 10

4) 5

Key: 3

Sol: $\bar{x} = 50$

$$\sum_{i=1}^{10} (x_i - \bar{x})^2 = 250$$

Maths - Question 36

Original paper page 14; original question and option regions

36. The variatnce of the first 50 even natural numbers is

1) $\frac{833}{4}$

2) 833

3) 437

4) $\frac{437}{4}$

Key: 2

Sol: 2, 4, 6,, 100

1, 2, 3,, 50

$$\bar{x} = \frac{\cancel{50} \times \frac{51}{2}}{\cancel{50}} = 25.5$$

$$\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$$

$$= \frac{1}{\cancel{50}} \times \frac{\cancel{50} \times 51 \times (101)}{6} - \left(\frac{51}{2}\right)^2$$

$$= \frac{51}{2} \left[\frac{101}{3} - \frac{51}{2} \right]$$

$$= \frac{\cancel{51}^{17}}{2} \left[\frac{49}{\cancel{6}_2} \right] = \frac{17 \times 49}{4} = \frac{833}{4}$$

$$v(2x) = 2^2 v(x) = 4 \times \frac{833}{4} = 833$$

Maths - Question 37

Original paper page 14; original question and option regions

37. 3 out of 6 vertices of a regular hexagon are chosen at a time at random. The probability that the triangle formed with these three vertices is an equilateral triangle, is

1) $\frac{1}{2}$

2) $\frac{1}{5}$

3) $\frac{1}{10}$

4) $\frac{1}{20}$

Key: 3

$$\text{Sol: } \frac{2}{{}^6C_3} = \frac{\cancel{2}}{\cancel{20}} = \frac{1}{10}$$

Maths - Question 38

Original paper page 14; original question and option regions

38. A speaks truth in 75% of the cases and B in 80% of the cases. Then the probability that their statements about an incident do not match, is

1) $\frac{7}{20}$

2) $\frac{3}{20}$

3) $\frac{2}{7}$

4) $\frac{5}{7}$

.....

Maths - Question 39

Original paper page 15; original question and option regions

39. If the mean and variance of a binomial distribution are 4 and 2 respectively, then the probability of 2 successes of that binomial variate X , is

1) $\frac{1}{2}$

2) $\frac{219}{256}$

3) $\frac{37}{256}$

4) $\frac{7}{64}$

Key: 4

Sol: $np = 4$; $npq = 2$

$$q = \frac{1}{2}, p = \frac{1}{2}, n = 8$$

$$P(X = 2) = {}^8C_2 q^6 p^2$$

$$= \frac{\cancel{28}}{\cancel{4} \times 2^6} = \frac{7}{64}$$

Maths - Question 40

Original paper page 15; original question and option regions

40. In a city 10 accidents take place in a span of 50 days. Assuming that the number of accidents flow the Poisson distribution, the probability that three or more accidents occur in a day, is

1) $\sum_{k=3}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!}, \lambda = 0.2$ 2) $\sum_{k=3}^{\infty} \frac{e^k \lambda^k}{k}, \lambda = 0.2$ 3) $1 - \sum_{k=3}^3 \frac{e^{-\lambda} \lambda^k}{k!}, \lambda = 0.2$ 4) $\sum_{k=3}^3 \frac{e^{-\lambda} \lambda^k}{k!}, \lambda = 0.2$

Key: 1

Sol: $\lambda = \frac{10}{50} = \frac{1}{5} = 0.2$

$$P(x \geq 3) = \sum_{k=3}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!}$$

Maths - Question 41

Original paper page 15; original question and option regions

41. Equation of the locus of the centroid of the triangle whose vertices are $(a \cos k, a \sin k)$, $(b \sin k, -b \cos k)$ and $(1, 0)$, where k is a parameter, is

1) $(1-3x)^2 + 9y^2 = a^2 + b^2$

2) $(3x-1)^2 + 9y^2 = 2a^2 + 2b^2$

3) $(3x+1)^2 + (3y)^2 = a^2 + b^2$

4) $(3x+1)^2 + (3y)^2 = 3a^2 + 3b^2$

Key: 1

$$\text{Sol: } (x, y) = \left(\frac{a \cos k + b \sin k + 1}{3}, \frac{a \sin k - b \cos k + 0}{3} \right)$$

$$a \cos k + b \sin k = 3x - 1$$

$$a \sin k - b \cos k = 3y$$

Squaring and adding

$$a^2 + b^2 = (3x-1)^2 + 9y^2$$

Maths - Question 42

Original paper page 16; original question and option regions

42. If the coordinate axes are rotated through an angle $\frac{\pi}{6}$ about the origin, then the transformed

equation of $\sqrt{3}x^2 - 4xy + \sqrt{3}y^2 = 0$ is

1) $\sqrt{3}y^2 + xy = 0$

2) $x^2 - y^2 = 0$

3) $\sqrt{3}y^2 - xy = 0$

4) $\sqrt{3}y^2 - 2xy = 0$

Key: 3

$$x = \frac{\sqrt{3}x - y}{2}$$

Sol: $y = \frac{x + y\sqrt{3}}{2}$

The transformed each line $\sqrt{3}y^2 - xy = 0$

Maths - Question 43

Original paper page 16; original question and option regions

43. If the lines $x + 3y - 9 = 0$, $4x + by - 2 = 0$, and $2x - y - 4 = 0$ are concurrent, then the equation of the line passing through the point $(b, 0)$ and concurrent with the given lines, is

- 1) $2x + y + 10 = 0$ 2) $4x - 7y + 20 = 0$ 3) $x - y + 5 = 0$ 4) $x - 4y + 5 = 0$

Key: 4

Sol: $\because$ Three lines are concurrent $\Rightarrow b = -5$

$\because$ point on intersection $= (-5, 0)$

point on concurrence $= (3, 2)$

$\therefore$ The equation of line passing through $(3, 2)$ & $(-5, 0)$ is $x - 4y + 5 = 0$

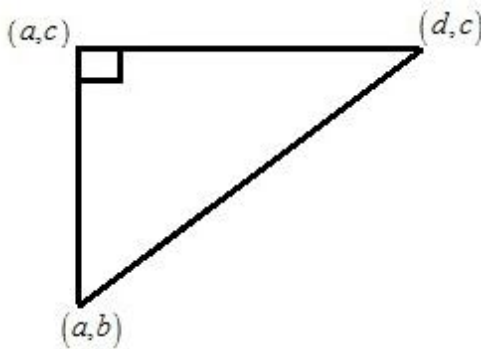
Maths - Question 44

Original paper page 16; original question and option regions

44. The midpoint of the line segment joining the centroid and the orthocentre of the triangle whose vertices are (a, b) , (a, c) and (d, c) , is

- 1) $\left(\frac{5a+d}{6}, \frac{b+5c}{6}\right)$ 2) $\left(\frac{a+5d}{6}, \frac{5b+c}{6}\right)$ 3) (a, c) 4) $(0, 0)$

Key: 1



Sol:

$$\text{Centroid, } G = \left(\frac{2a+d}{3}, \frac{2c+b}{3}\right)$$

$$\text{Orthocentre } O = (a, c)$$

$$\text{Midpoint} = \left(\frac{5a+d}{6}, \frac{5c+b}{6}\right)$$

Maths - Question 45

Original paper page 17; original question and option regions

45. The distance from the origin to the image of (1, 1) with respect to the line $x+y+5=0$ is

1) $7\sqrt{2}$

2) $3\sqrt{2}$

3) $6\sqrt{2}$

4) $4\sqrt{2}$

Key: 3

Sol: Image (-6, -6)

Distance from origin $\sqrt{36+36} = \sqrt{72} = 6\sqrt{2}$

Maths - Question 46

Original paper page 17; original question and option regions

46. The equation of the pair of lines joining the origin to the points of intersection of $x^2 + y^2 = 9$ and $x + y = 3$, is

- 1) $x^2 + (3 - y)^2 = 9$ 2) $(3 + y)^2 + y^2 = 9$ 3) $x^2 - y^2 = 9$ 4) $xy = 0$

Key: 4

$$\text{Sol: } x^2 + y^2 - 9\left(\frac{x+y}{3}\right)^2 = 0$$

$$\Rightarrow x^2 + y^2 - (x^2 + y^2 + 2xy) = 0$$

$$\Rightarrow xy = 0$$

Maths - Question 47

Original paper page 17; original question and option regions

47. The orthocentre of the triangle formed by the lines $x + y = 1$ and $2y^2 - xy - 6x^2 = 0$ is

1) $\left(\frac{4}{3}, \frac{4}{3}\right)$

2) $\left(\frac{2}{3}, \frac{2}{3}\right)$

3) $\left(\frac{2}{3}, \frac{-2}{3}\right)$

4) $\left(\frac{4}{3}, \frac{-4}{3}\right)$

Key: 1

Sol: $x + y = 1$

$$2y^2 - 4xy + 3xy - 6x^2 = 0$$

$$2y(y - 2x) + 3x(y - 2x) = 0$$

$$(y - 2x)(3x + 2y) = 0$$

$$y - 2x = 0$$

$$3x + 2y = 0$$

$$x - y = 0$$

$$\left(\frac{4}{3}, \frac{4}{3}\right)$$

Maths - Question 48

Original paper page 17; original question and option regions

48. Let L be the line joining the origin to the point of intersection of the lines represented by $2x^2 - 3xy - 2y^2 + 10x + 5y = 0$. If L is perpendicular to the line $kx + y + 3 = 0$, then k=

1) $\frac{1}{2}$

2) $\frac{-1}{2}$

3) -1

4) $\frac{1}{3}$

Key: 2

Sol: $2x^2 - 3xy - 2y^2 + 10x\left(\frac{kx+y}{-3}\right) + 5y\left(\frac{kx+y}{-3}\right) = 0$

$$\frac{-10k}{3} - \frac{5}{3} = 0$$

$$2k + 1 = 0$$

$$k = -\frac{1}{2}$$

Maths - Question 49

Original paper page 18; original question and option regions

49. A circle $S=0$ with radius $\sqrt{2}$ touches the line $x + y - z = 0$ at $(1, 1)$. Then the length of the tangent drawn from the point $(1, 2)$ to $S=0$ is

1) 1

2) $\sqrt{2}$

3) $\sqrt{3}$

4) 2

Key: Add score

Sol: $x + y - 2 = 0$

$$(x^2 + y^2 - 2x - 2y + 2) + 2\lambda(x + y - 2) = 0$$

$$x^2 + y^2 + 2x(\lambda - 1) + 2y(\lambda - 1) + 2 - 4\lambda = 0$$

$$\lambda - 11^2 + (\lambda - 1)^2 - 2 + 4\lambda = 2$$

$$2\lambda^2 = 2$$

$$\lambda^2 = 1$$

$$\lambda = \pm 1$$

Maths - Question 50

Original paper page 18; original question and option regions

50. The normal drawn at $P(-1, 2)$ on the circle $x^2 + y^2 - 2x - 2y - 3 = 0$ meets the circle at another point Q. Then the coordinates of Q are

1) (3, 0)

2) (-3, 0)

3) (2, 0)

4) (-2, 0)

Key: 1

Sol: $P(-1, 2)$ $C = (1, 1)$

$$Q = 2C - P$$

$$= (2, 2) - (-1, 2)$$

$$= (3, 0)$$

Maths - Question 51

Original paper page 18; original question and option regions

51. If the lines $kx + 2y - 4 = 0$ and $5x - 2y - 4 = 0$ are conjugate with respect to the circle

$x^2 + y^2 - 2x - 2y + 1 = 0$, then $k =$

1) 0

2) 1

3) 2

4) 3

Key: 2

Sol: $(5(k) - 4)1 = (k - 2)(-1)$

$$5k - 4 = -k + 2$$

$$6k = 6$$

$$k = 1$$

Maths - Question 52

Original paper page 18; original question and option regions

52. The angle between the tangents drawn from the origin to the circle $x^2 + y^2 + 4x - 6y + 4 = 0$ is

1) $\tan^{-1}\left(\frac{5}{13}\right)$

2) $\tan^{-1}\left(\frac{5}{12}\right)$

3) $\tan^{-1}\left(\frac{12}{5}\right)$

4) $\tan^{-1}\left(\frac{13}{5}\right)$

Key: 3

$$\text{Sol: } \tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}} = \frac{\sqrt{4+9-4}}{2} = \frac{3}{2} \quad \left| \quad \tan \theta = \frac{2\left(\frac{3}{2}\right)}{1 - \frac{9}{4}} = \frac{3 \times 4}{-5} = \frac{12}{5} \right.$$

$$\theta = \tan^{-1}\left(\frac{12}{5}\right)$$

Maths - Question 53

Original paper page 19; original question and option regions

53. If the angle between the circles $x^2 + y^2 - 2x - 4y + c = 0$ and $x^2 + y^2 - 4x - 2y + 4 = 0$ is 60° , then c is equal to

1) $\frac{3 \pm \sqrt{5}}{2}$

2) $\frac{6 \pm \sqrt{5}}{2}$

3) $\frac{9 \pm \sqrt{5}}{2}$

4) $\frac{7 \pm \sqrt{5}}{2}$

Key: 4

Sol: $C_1 = (1, 2)$ $C_2 = (2, 1)$ $d = \sqrt{1+1}$ $r_1 = \sqrt{5-c}$ $r_2 = 1$

$$\cos 60^\circ = \frac{d^2 - r_1^2 - r_2^2}{2r_1r_2}$$

$$2 = \frac{2 - (5-c) - 1}{2\sqrt{5-c}}$$

$$\sqrt{5-c} = c - 4$$

$$5-c = (c-4)^2$$

$$5-c = c^2 - 8c + 16$$

$$c^2 - 7c + 11 = 0$$

$$c = \frac{7 \pm \sqrt{49-44}}{2}$$

$$c = \frac{7 \pm \sqrt{5}}{2}$$

Maths - Question 54

Original paper page 19; original question and option regions

54. A circle S cuts three circles $x^2 + y^2 - 4x - 2y + 4 = 0$, $x^2 + y^2 - 2x - 4y + 1 = 0$, and $x^2 + y^2 + 4x + 2y + 1 = 0$ orthogonally. Then the radius of S is

1) $\sqrt{\frac{29}{8}}$

2) $\sqrt{\frac{28}{11}}$

3) $\sqrt{\frac{29}{7}}$

4) $\sqrt{\frac{29}{5}}$

Key: 1

Sol: $2x - 2y - 3 = 0$

$$6x + 6y = 0 \Rightarrow x + y = 0$$

$$x = \frac{3}{4}, y = \frac{3}{4}$$

$$\left(\frac{3}{4}, \frac{3}{4}\right)$$

$$r = \sqrt{\frac{9}{16} + \frac{9}{16} - \frac{12}{4} + \frac{6}{4} + 4}$$

$$= \sqrt{\frac{18 - 48 + 24 + 64}{4}}$$

$$= \frac{\sqrt{58}}{4}$$

$$= \sqrt{\frac{58}{16}}$$

$$= \sqrt{\frac{29}{8}}$$

Maths - Question 55

Original paper page 20; original question and option regions

55. The distance between the vertex and the focus of the parabola $x^2 - 2x + 3y - 2 = 0$ is

1) $\frac{4}{5}$

2) $\frac{3}{4}$

3) $\frac{1}{2}$

4) $\frac{5}{6}$

Key: 2

Sol: $(x-1)^2 + 3y - 3 = 0$

$$(x-1)^2 = -3(y-1)$$

$$4a = 3$$

$$a = \frac{3}{4}$$

Maths - Question 56

Original paper page 20; original question and option regions

56. If (x_1, y_1) and (x_2, y_2) are the end points of a focal chord of the parabola $y^2 = 5x$, then

$$4x_1x_2 + y_1y_2 =$$

1) 25

2) 5

3) 0

4) $\frac{5}{4}$

Key: 3

Sol: $4a = 5$

$$4(at_1^2 \cdot t_2^2) + (2at_1)(2at_2)$$

$$= 4a^2(t_1t_2)^2 + 4a^2(t_1t_2)$$

$$= 4a^2 - 4a^2$$

$$= 0$$

Maths - Question 57

Original paper page 20; original question and option regions

57. The distance between the foci of the ellipse $x = 3\cos\theta, y = 4\sin\theta$ is

1) $2\sqrt{7}$

2) $7\sqrt{2}$

3) $\sqrt{7}$

4) $3\sqrt{7}$

Key: 1

Sol: $\frac{x^2}{9} + \frac{y^2}{16} = 1$

$$b = 4, a = 3$$

$$e = \sqrt{\frac{16-9}{16}} = \frac{\sqrt{7}}{4}$$

$$be = \sqrt{7}$$

$$2be = 2\sqrt{7}$$

Maths - Question 58

Original paper page 20; original question and option regions

58. The equation of the latus recta of the ellipse $9x^2 + 25y^2 - 36x + 50y - 164 = 0$ are

- 1) $x - 4 = 0, x + 2 = 0$ 2) $x - 6 = 0, x + 2 = 0$ 3) $x + 6 = 0, x - 2 = 0$ 4) $x + 4 = 0, x + 5 = 0$

Key: 2

Sol: $9x^2 + 25y^2 - 36x + 50y - 164 = 0$

Maths - Question 59

Original paper page 21; original question and option regions

59. The values of m for which the lines $y = mx + 2$ becomes a tangent to the hyperbola $4x^2 - 9y^2 = 36$ is

1) $\pm \frac{2}{3}$

2) $\pm \frac{2\sqrt{2}}{3}$

3) $\pm \frac{8}{9}$

4) $\pm \frac{4\sqrt{2}}{3}$

Key: 2

Sol: $\frac{x^2}{9} - \frac{y^2}{4} = 1$

$y = mx + 2$

$$9(m^2) - 4 = 4$$

$$m^2 = \frac{8}{9} \quad m = \pm \frac{2\sqrt{2}}{3}$$

Maths - Question 60

Original paper page 21; original question and option regions

60. The harmonic conjugate of (2, 3, 4) with respect to the points (3, -2, 2), (6, -17, -4) is

1) $\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right)$

2) $\left(\frac{18}{5}, -5, \frac{4}{5}\right)$

3) $\left(\frac{-18}{5}, \frac{5}{4}, \frac{4}{5}\right)$

4) $\left(\frac{18}{5}, -5, \frac{-4}{5}\right)$

Key: 2

Sol: (2,3,4) divides A (3,-2,2) and B (6,-17,-4) to the ratio 1:-4

∴ The harmonic conjugate at (2,3,4)

divide $\overline{AB}$ is the ratio 1:4

$$\therefore \text{Ratio} = \left(\frac{18}{5}, -5, \frac{4}{5}\right)$$

Maths - Question 61

Original paper page 21; original question and option regions

61. If a line makes angles α, β, γ and δ with the four diagonals of a cube, then the value of $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta$ is

1) $\frac{4}{3}$

2) $\frac{8}{3}$

3) $\frac{7}{3}$

4) $\frac{5}{3}$

Key: 2

Maths - Question 62

Original paper page 22; original question and option regions

62. If the plane $56x + 4y + 9z = 2016$ meets the coordinate axes in A, B, C then the centroid of the triangle ABC is

- 1) $(12, 168, 224)$ 2) $(12, 168, 112)$ 3) $\left(12, 168, \frac{224}{3}\right)$ 4) $\left(12, -168, \frac{224}{3}\right)$

Key: 3

Sol: $\frac{56x}{2016} + \frac{4y}{2016} + \frac{9z}{2016} = 1$

$$\frac{x}{36} + \frac{y}{504} + \frac{z}{224} = 1$$

$A(36, 0, 0)$ $B(0, 504, 0)$ $C(0, 0, 224)$

$$\text{Centroid} = \left(\frac{36}{3}, \frac{504}{3}, \frac{224}{3}\right)$$

Maths - Question 63

Original paper page 22; original question and option regions

63. The value(s) of x for which the function

$$f(x) = \begin{cases} 1-x & , x < 1 \\ (1-x)(2-x) & , 1 \leq x \leq 2 \\ 3-x & , x > 2 \end{cases}$$

fails to be continuous is (are)

1) 1

2) 2

3) 3

4) All real numbers

Key: 2

Sol: $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$
 $f(x)$ is discontinuous at $x = 2$

$$2^x - 2^x = 2^x - 1$$

Maths - Question 64

Original paper page 22; original question and option regions

64. $\lim_{x \rightarrow 0} \frac{3^x - 3^{x-2} - 1}{x^2} =$

1) $(\log_e 2) \log_e 3$

2) $\log_e 5$

3) $\log_e 6$

4) 0

Key: 1

$$\lim_{x \rightarrow 0} \frac{3^x (2^x - 1) - 1(2^x - 1)}{x^2}$$

$$\begin{aligned} \text{Sol: } &= \lim_{x \rightarrow 0} \frac{(3^x - 1)}{x} \times \lim_{x \rightarrow 0} \left(\frac{2^x - 1}{x} \right) \\ &= \log_e 3 \times \log_e 2 \end{aligned}$$

Maths - Question 65

Original paper page 23; original question and option regions

65. Define $f(x) = \begin{cases} 2x + b, & x < 1 \\ 1 + c, & x \geq 1 \end{cases}$. If $f(x)$ is differentiable at $x=1$, then $(b-c) =$

1) -2

2) 0

3) 1

4) 2

Key: 1

Sol: $f(x)$ is continuous and differentiable at $x=1$

$$\Rightarrow 1 + b + c = 1 \Rightarrow b + c = 0$$

$$f'(x) = \begin{cases} 2, & x < 1 \\ c, & x \geq 1 \end{cases}$$

$$\Rightarrow 2 = c, \quad b = -2, \quad c = 2$$

$$\therefore b - c = -2 - 2 = -4$$

Maths - Question 66

Original paper page 23; original question and option regions

66. If $x = a$ is a root of multiplicity two of a polynomial equation $f(x) = 0$, then

1) $f^1(a) = f^{11}(a) = 0$

2) $f^{11}(a) = f(a) = 0$

3) $f^1(a) \neq 0 \neq f^{11}(a)$

4) $f(a) = f^1(a) = 0; f^{11}(a) \neq 0$

Key: 4

Sol: Conceptual

$$f(a) = f^1(a) = 0; f^{11}(a) \neq 0$$

Ans,

Maths - Question 67

Original paper page 23; original question and option regions

67. If $y = \log_2(\log_2 x)$, then $\frac{dy}{dx} =$

1) $\frac{\log_e 2}{x \log_e x}$

2) $\frac{1}{\log_e (2x)^x}$

3) $\frac{1}{(x \log_e x) \log_e 2}$

4) $\frac{1}{x(\log_e x)^2}$

Key: 3

Sol: $y = \log_2(\log_2 x)$

$$\frac{dy}{dx} = \frac{1}{\log_2 x \times x (\log_e 2)^2}$$

$$= \frac{1}{\frac{\log_e x}{(\log_e 2)} \times x (\log_e 2)^2}$$

$$= \frac{1}{(x \cdot \log_e x) \times \log_e 2}$$

Maths - Question 68

Original paper page 23; original question and option regions

68. The angle of intersection between the curves $y^2 + x^2 = a^2\sqrt{2}$ and $x^2 - y^2 = a^2$, is

1) $\frac{\pi}{3}$

2) $\frac{\pi}{4}$

3) $\frac{\pi}{6}$

4) $\frac{\pi}{12}$

Key: 2

Sol: $x^2 + y^2 = \sqrt{2}a^2$

$$x^2 - y^2 = a^2$$

Maths - Question 69

Original paper page 24; original question and option regions

69. If $A > 0$, $B > 0$ and $A + B = \frac{\pi}{3}$, then the maximum value of $\tan A \tan B$ is

1) $\frac{1}{\sqrt{3}}$

2) $\frac{1}{3}$

3) $\frac{1}{2}$

4) $\sqrt{3}$

Key: 2

Sol: $B = \frac{\pi}{3} - A$

$$f(A) = \tan A \tan \left(\frac{\pi}{3} - A \right)$$

$$f'(A) = 0 \Rightarrow A = \frac{\pi}{6}$$

$$\text{Max. value} = \tan \left(\frac{\pi}{6} \right) \tan \left(\frac{\pi}{6} \right)$$

$$= \frac{1}{3}$$

Maths - Question 70

Original paper page 25; original question and option regions

70. The equation of the common tangent drawn to the curves $y^2 = 8x$ and $xy = -1$ is

1) $y = 2x + 1$

2) $2y = x + 6$

3) $y = x + 2$

4) $3y = 8x + 2$

Key: 3

Sol: $y^2 = 8x, xy = -1$

$$y = mx + \frac{2}{m}$$

$$x\left(mx + \frac{2}{m}\right) = -1$$

$$x^2m + \frac{2}{m}x + 1 = 0$$

$$\Delta = 0$$

$$\frac{4}{m^2} - 4m = 0$$

$$m^3 = 1$$

$$m = 1$$

$$y = x + 2$$

Maths - Question 71

Original paper page 25; original question and option regions

71. Suppose $f(x) = x(x+3)(x-2)$, $x \in [-1, 4]$. Then a value of c in $(-1, 4)$ satisfying $f'(c) = 10$ is

1) 2

2) $\frac{5}{2}$

3) 3

4) $\frac{7}{2}$

Key: 1

Sol: $f(x) = x^3 + x^2 - 6x$

$$f'(x) = 3x^2 + 2x - 6 = 10$$

$$3x^2 + 2x - 16 = 0$$

$$3x^2 + 8x - 6x - 16 = 0$$

$$x(3x+8) - 2(3x+8) = 0$$

$$x = 2, \frac{-8}{3}$$

$$x = 2$$

Maths - Question 72

Original paper page 25; original question and option regions

72. If $\int x^3 e^{5x} dx = \frac{e^x}{5^4} (f(x)) + c$, then $f(x) =$

1) $\frac{x^3}{5} - \frac{3x^2}{5^2} + \frac{6x}{5^3} - \frac{6}{5^4}$

2) $5x^3 - 5^2 x^2 + 5^3 x - 6$

3) $5^2 x^3 - 15x^2 + 30x - 6$

4) $5^3 x^3 - 75x^2 + 30x - 6$

Key: 4

Maths - Question 73

Original paper page 26; original question and option regions

73. $\int \frac{1}{(x^2 + 2x + 2)^2} dx =$

1) $\frac{x^2 + 2}{x^2 + 2x + 2} - \frac{1}{2} \tan^{-1}(x+1) + c$

2) $\frac{x^2 + 2}{2(x^2 + 2x + 2)} - \frac{1}{2} \tan^{-1}(x-1) + c$

3) $\frac{x^2 - 2}{4(x^2 + 2x + 2)} - \frac{1}{2} \tan^{-1}(x+1) + c$

4) $\frac{2(x-1)}{(x^2 + 2x + 2)} + \frac{1}{2} \tan^{-1}(x+1) + c$

(C is an arbitrary constant)

Key: 3

Sol: Put $(x+1) = \tan \theta$

$$dx = \sec^2 \theta d\theta$$

$$\begin{aligned} \therefore I &= \int \frac{(\tan \theta - 1) \cdot \sec^2 \theta}{\sec^4 \theta} d\theta \\ &= -\frac{1}{4} \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right) - \frac{1}{4} \left(\frac{2 \tan \theta}{1 + \tan \theta} \right) - \frac{1}{2} \theta + C \\ &= \frac{1}{4} \frac{(2^2 - 2)}{(x^2 + 2x + 2)} - \frac{1}{2} \tan^{-1}(x+1) + C \end{aligned}$$

Maths - Question 74

Original paper page 26; original question and option regions

74. If $\int \log(a^2 + x^2) dx = h(x) + c$, then $h(x) =$

1) $x \log(a^2 + x^2) + 2 \tan^{-1}\left(\frac{x}{a}\right)$

2) $x^2 \log(a^2 + x^2) + x + a \tan^{-1}\left(\frac{x}{a}\right)$

3) $x \log(a^2 + x^2) - 2x + 2a \tan^{-1}\left(\frac{x}{a}\right)$

4) $x^2 \log(a^2 + x^2) + 2x - a^2 \tan^{-1}\left(\frac{x}{a}\right)$

Key: 3

Sol: $\int \frac{\log(a^2 + x^2)}{f(x)} \cdot \frac{1}{g(x)} dx$

$$= \log(a^2 + x^2) \cdot x - \int \frac{1}{(a^2 + x^2)} 2x^2 dx$$

$$= \log(a^2 + x^2) \cdot x - 2 \int \frac{a^2 + x^2 - a^2}{(a^2 + x^2)} dx$$

$$x \cdot \log(a^2 + x^2) + 2x - a^2 \tan^{-1}(x/a) + C$$

Maths - Question 75

Original paper page 27; original question and option regions

75. For $x > 0$, if $\int (\log x)^5 dx =$

$$x \left[A(\log x)^5 + B(\log x)^4 + C(\log x)^3 + D(\log x)^2 + E(\log x) + F \right] + \text{constant, then}$$

$$A + B + C + D + E + F =$$

1) -44

2) -42

3) -40

4) -36

Key: 1

Sol: $\int (\log x)^5 dx$

$$= x \left[(\log x)^5 - (\log x)^4 + 20(\log x)^3 - 60(\log x)^2 + 120(\log x) - 120 \right] + C$$

$$A = 1, B = -5, C = 20, D = -60, E = 120, F = -120$$

$$A + B + C + D + E + F = -44$$

Maths - Question 76

Original paper page 27; original question and option regions

76. The area included between the parabola $y = \frac{x^2}{4a}$ and the curve $y = \frac{8a^3}{(x^2 + 4a^2)}$ is

- 1) $a^2 \left(2\pi + \frac{2}{3} \right)$ 2) $a^2 \left(2\pi - \frac{8}{3} \right)$ 3) $a^2 \left(\pi + \frac{4}{3} \right)$ 4) $a^2 \left(2\pi - \frac{4}{3} \right)$

Key: 4

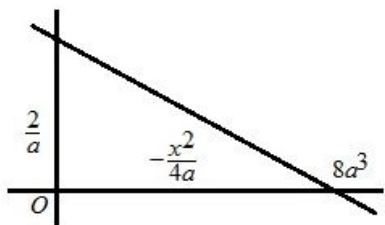
Sol: $x = 4ay, y = \frac{8a^3}{(x^2 + 4a^2)}$

$$x^4 + 4a^2x^2 = 32a^4$$

$$x^4 + 4a^2x^2 - 32a^4 = 0$$

$$(x^2 + 8a^2)(x^2 - 4a^2) = 0$$

$$x = \pm 2a$$



$$A = \int_{-2a}^{2a} \left(\frac{8a^3}{x^2 + 4a^2} - \frac{x^2}{4a} \right) dx$$

$$= 16a^3 \times \frac{1}{2a} \tan^{-1} \left(\frac{x}{2a} \right) \Big|_{-2a}^{2a} - \frac{1}{6a} x^3 \Big|_{-2a}^{2a}$$

$$8a^2 \left(\frac{\pi}{4} \right) - \frac{1}{6a} (8a^3)$$

$$= 2a^2\pi - \frac{4}{3}a^2$$

$$= a^2 \left(2\pi - \frac{4}{3} \right)$$

Maths - Question 77

Original paper page 28; original question and option regions

77. By the definition of the definite integral, the value of

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2 - 1}} + \frac{1}{\sqrt{n^2 - 2^2}} + \dots + \frac{1}{\sqrt{n^2 - (n-1)^2}} \right) \text{ is equal to}$$

1) π

2) $\frac{\pi}{2}$

3) $\frac{\pi}{4}$

4) $\frac{\pi}{6}$

Key: 2

$$\text{Sol: } \lim_{x \rightarrow \infty} \left(\sum_{r=1}^{n-1} \frac{1}{\sqrt{n^2 - r^2}} \right)$$

$$\lim_{x \rightarrow \infty} \left(\frac{1}{n} \right) \sum_{r=1}^{n-1} \frac{1}{\sqrt{1 - (r/n)^2}}$$

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$$

$$= (\sin^{-1} x)_0^1$$

$$= \pi / 2$$

$$\left[\sin^{-1} x + \frac{\pi}{4} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

Maths - Question 78

Original paper page 28; original question and option regions

78. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{2 - \cos 2x} \, dx =$

1) $\frac{8\pi\sqrt{3}}{5}$

2) $\frac{2\pi\sqrt{3}}{9}$

3) $\frac{4\pi^2\sqrt{3}}{9}$

4) $\frac{\pi^2}{6\sqrt{3}}$

Key: 4

Sol: $I = \frac{\pi}{4} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2 - \cos 2x} \, dx$ put $\tan x = t$ $dx = \frac{dt}{1+t^2}$

$$= 2 \times \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \frac{1}{2 - \cos 2x} \, dx \quad \cos 2x = \frac{1-t^2}{1+t^2}$$

$$= \frac{\pi}{2} \int_0^1 \frac{1}{3t^2 + 1} \, dt$$

$$= \frac{\pi}{2\sqrt{3}} \times \frac{\pi}{3} = \frac{\pi}{6\sqrt{3}}$$

Maths - Question 79

Original paper page 29; original question and option regions

79. The solution of the differential equation $(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$, is

1) $x e^{\tan^{-1} y} = \tan^{-1} y + c$

2) $x e^{2 \tan^{-1} y} = e^{-\tan^{-1} y} + c$

3) $2x e^{\tan^{-1} y} = e^{2 \tan^{-1} y} + c$

4) $x^2 e^{\tan^{-1} y} = 4e^{2 \tan^{-1} y} + c$

(C is an arbitrary constant)

Key: 3

Sol: $\frac{dx}{dy} = \frac{e^{\tan^{-1} y} - x}{(1 + y^2)}$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{(1 + y^2)} = \frac{e^{\tan^{-1} y}}{(1 + y^2)}$$

This is L.D.E is x

$$\text{I.F} = \frac{e^{\tan^{-1} y}}{(1 + y^2)}$$

Solution is

$$x \cdot e^{\tan^{-1} y} = \int \frac{e^{\tan^{-1} y}}{(1 + y^2)^2} \cdot e^{\tan^{-1} y} y \cdot dy$$

$$= \frac{(e^{\tan^{-1} y})^2}{2} + C$$

$$\Rightarrow 2x \cdot e^{\tan^{-1} y} = e^{2 \tan^{-1} y} + C$$

Maths - Question 80

Original paper page 29; original question and option regions

80. The solution of the differential equation $(2x - 4y + 3)\frac{dy}{dx} + (x - 2y + 1) = 0$ is

1) $\log((2x - 4y) + 3) = x - 2y + c$

2) $\log[2(2x - 4y) + 3] = 2(x - 2y) + c$

3) $\log[2(x - 2y) + 5] = 2(x + y) + c$

4) $\log[4(x - 2y) + 5] = 4(x + 2y) + c$

(C is an arbitrary constant)

Key: 4

Sol: Put $x - 2y = z$

$$1 - 2\frac{dy}{dx} = \frac{dz}{dx}$$

$$2\frac{dy}{dx} = 1 - \frac{dz}{dx}$$

$$\frac{dy}{dx} = \frac{1}{2}\left(1 - \frac{dz}{dx}\right)$$